NeverMiss.ai — Phase 2 Architecture & Workflow

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1. Executive Overview & Scope

NeverMiss.ai requires a **Self-Improving Fleet Management System** for its ElevenLabs voice agents. The objective of Phase 2 is to monitor live call performance from a single pane of glass, automate 100% post-call quality evaluations, and establish a Human-in-the-Loop (HITL) gatekeeper for AI-suggested prompt revisions.

Component	Description	Target SLA / Tech Stack
Webhook Ingestion	Stream 100% ElevenLabs post-call webhooks to n8n listener	n8n on p330 / Cloudflare Tunnel
AI Evaluation Engine	Score call transcripts against custom quality rubrics	Gemini 2.0 Flash Multimodal
Database Layer	Multi-tenant storage with Row-Level Security (RLS)	Azure PostgreSQL Flexible Server (pgvector)
HITL Gatekeeper	Admin review queue for approving AI prompt revisions	Next.js 16 + Tailwind CSS
Frontend Deploy	Automated GitHub Actions PR deployment pipeline	Azure Static Web Apps (spf-admin repo)

2. System Logic & Workflow Pipeline

The system operates on an automated, self-correcting feedback loop:

- **Step 1 (Ingestion):** ElevenLabs completes a voice call and fires a post-call webhook payload to https://n8n.rivasautomations.com/webhook/elevenlabs.
- **Step 2 (Evaluation):** n8n routes the transcript and audio metadata to Gemini 2.0 Flash to score against active Eval criteria (e.g. Email Capture, Tone, Escalation rules).
- **Step 3 (Storage):** Evaluation results, structured transcripts, and vector embeddings are stored in Azure PostgreSQL Flexible Server under strict tenant isolation.
- **Step 4 (HITL Review):** If a call fails an Eval, an LLM diagnoses the root cause, drafts a system prompt fix, and pushes it to the Dashboard HITL Queue.
- **Step 5 (Self-Correction):** Upon one-click human approval in the dashboard, the updated system prompt is pushed directly back to ElevenLabs via API.

3. Admin Dashboard Screen Map

Screen Name	Core Functionality & Deliverable Scope
1. Dashboard Overview	Eval pass rates, p50 turn latency, booking webhook latency, live call metrics, active fleet status.
2. Voice Agents Fleet	Agent cards, ElevenLabs configuration (Voice ID, greeting script, Twilio numbers, calendar mappings).
3. Evals & Rubrics ■	Rubric builder (greeting rules, lead qualification, spam filter), per-agent rubric assignment, retrofitting unconfigured agents.
4. HITL Review Queue ■	Incident queue for failed calls. Side-by-side prompt diff viewer with one-click 'Approve & Deploy' action.
5. Call Logs & Transcripts	Searchable transcript history, audio waveform visualizer, audio playback, per-call evaluation breakdown.
6. Settings & Integrations	Azure PostgreSQL connection details, n8n webhook configuration, Google Calendar & CRM mappings (ServiceTitan, Jobber).

4. Azure PostgreSQL Database & Security Protocol

A multi-tenant database DDL schema has been generated in nevermiss_schema.sql. It includes Row-Level Security (RLS) policies on all tables keyed on tenant_id to guarantee client data isolation. Required Azure Extension: pgvector (for semantic vector search over call transcripts).

- **Primary Tables:** tenants, voice_agents, calls, transcripts, recordings, eval_rubrics, eval_criteria, eval_results, prompt_revisions, appointments, customers.
- **Secret Management:** API keys and database credentials will be stored exclusively in Azure Key Vault / n8n environment secrets.